

Experiment 4: Step-Up DC-DC Converter

Part 1: Open-Loop Cascaded Boost DC-DC Converter

ECEN 5517-012

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BENCH 5

I.Introduction

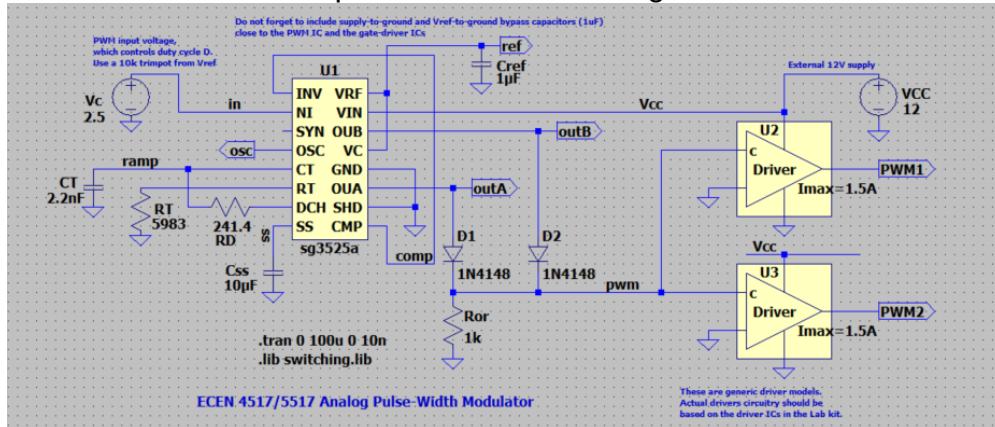
The goal of this experiment is to boost the battery voltage to a high enough DC to be able to invert to 120 AC. We accomplish this with two distinct boost stages, complemented by a PWM generator and two of the same gate drivers as used for the buck converter. The first part of this experiment will use a trimpot to implement our duty cycle control.

II.Experimental Procedure and Results

a. Floor-Planning ☒

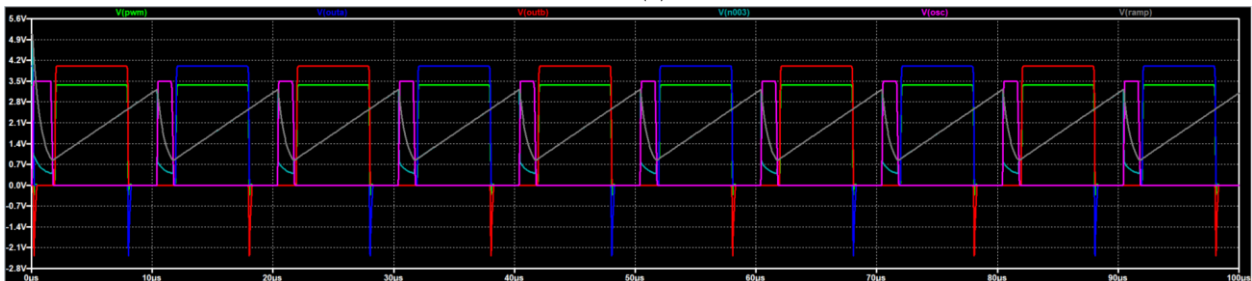
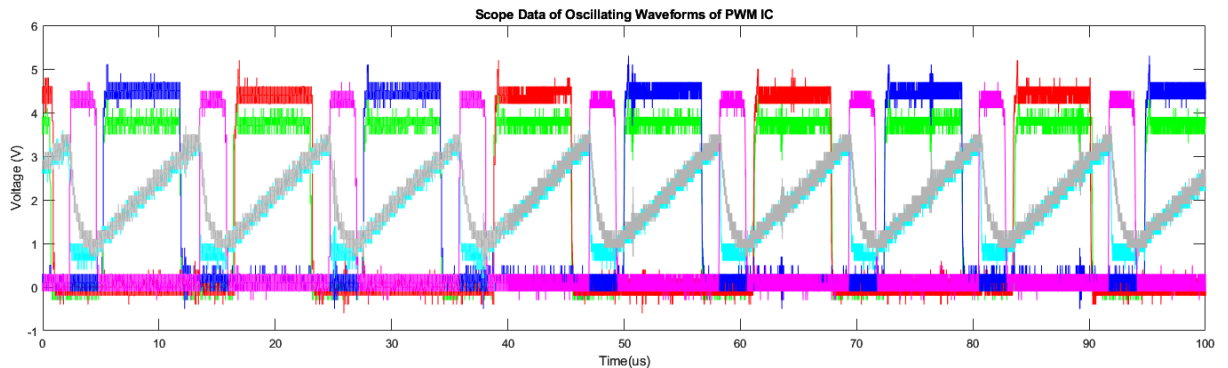
b. Analog PWM Circuit

i. Complete PWM Circuit Diagram



ii. Labeled Waveforms and Measured DC Values for all 16 pins of the PWM IC and PWM Circuit Output Vpwm.

WaveForm Signals (OSC, CT (ramp), OUTA, OUTB, DCH, and PWM) vs Simulation



PWM – Green, OUTB – Red, OUTA – Blue, DCH – Cyan, OSC – Magenta, CT (Ramp) - Grey

DC Pin Measurements

	INV	NI	SYN	RT	SS	VRF	VIN	VC	GND	SHD	COMP
Measured	2.799	2.799	0.022	3.735	4.907	5.156	12.13	5.156	0	0	2.798
Simulation	2.498	2.500	0	3.640	5.075	5.123	12.00	5.123	0	0	2.498

iii. Comparison to Simulation Results

The input voltage of our real-world measurements will skew results slightly as it messes with the bias points of the analog active circuits generating the sawtooth waveforms. The required input voltage to generate a 60% duty cycle in the simulation is only 2.5V compared to 2.8V in real-world testing. The output high of the PWM is also higher than the 3.4V of the simulation, closer to 4V with our measured V_c (presumably through a bandgap voltage reference that is not resilient to swings in VDD). The switching frequency of the simulation is 99.93 kHz with a duty cycle of exactly 60.2%. The real-world duty cycle is 60.22% at a switching frequency of exactly 89.98 kHz as well, so a bit lower than expected. If we had taken more care to have a clean 12V input voltage, our values would've been far more similar, but they nonetheless indicate a functioning and spec-meeting PWM circuit.

c. Inductor Design and Construction

i. Documented Inductor Design and Measured Inductances

The input inductor was designed for a ripple of 10% at a switching frequency of 100 kHz so that we could deal with relatively low power levels without entering DCM. It was also designed to assume a peak power of 85W from the source. The input current from the battery assuming a 12V input voltage is $85W/12V = 7.083A$. The ripple of this is roughly 0.71 A for 10% constraint. We used the following equation derived from the boost converter to determine our inductance of **65 μH** :

$$L = \frac{V_{batt}DT_s}{2\Delta i_L}$$

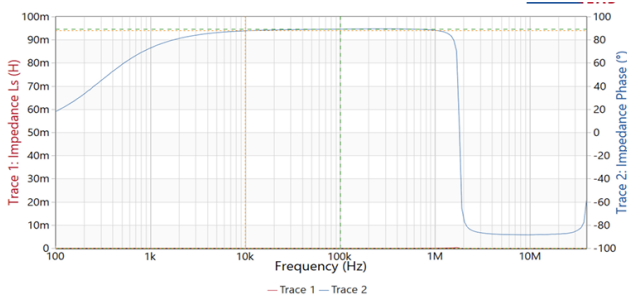
We can use the PQ 26/25 core since it's the smallest we have with a suitable K_g . Using the calculated inductance above, a maximum current through the first boost stage of 7.8 A, a B_{max} of 0.35 T, a typical K_u of 0.5, and the given bobbin parameters, we calculated the rest of the inductor's parameters using the equations shown below.

$$n \geq \frac{LI_{max}}{B_{max}A_c} = 12.2 \rightarrow \mathbf{13 \text{ turns}}$$

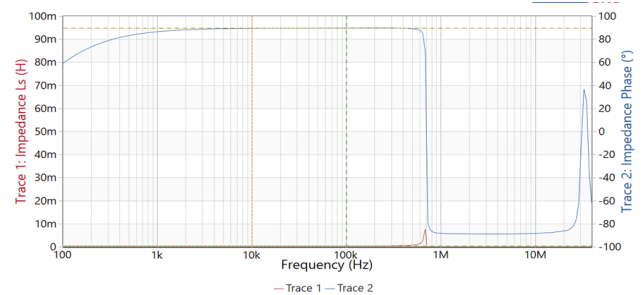
$$l_g = \frac{\mu_0 A_c n^2}{L} = \mathbf{15 \text{ mils}}$$

$$\frac{K_u W_a}{n} = .0206 \text{ cm}^2 \rightarrow \mathbf{16 \text{ AWG}}$$

The second boost stage inductor is calculated with the same L_s equation, except the input of the second boost stage is 42.43 V instead of 12. This gives a new current of about 2 A. We also selected a higher current ripple of 15% to improve the manufacturability of the second inductor while also trying to prevent saturation, giving a value of **400 μH** . Using the same equations and constants as above, the second inductor will have **24 turns of 18 AWG** wire and an **8 mil** airgap. Both inductors needed some adjustments to the airgap, and the final constructed inductors were **64.7 μH** and **401.5 μH** with **DC resistances of 10.4 and 26.5 m Ω** , respectively.

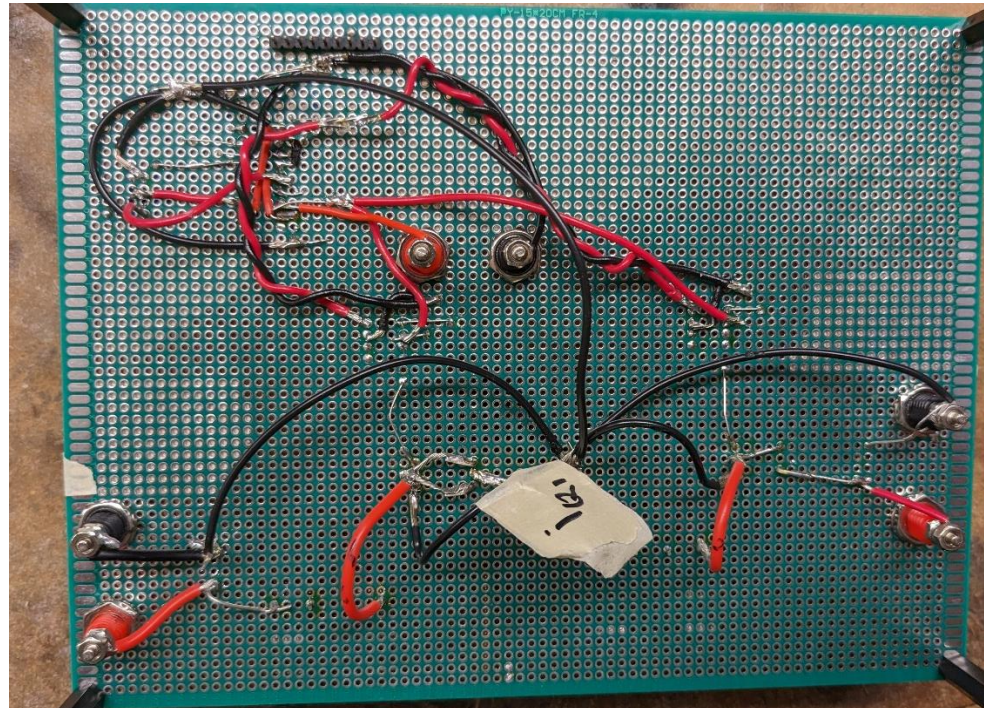
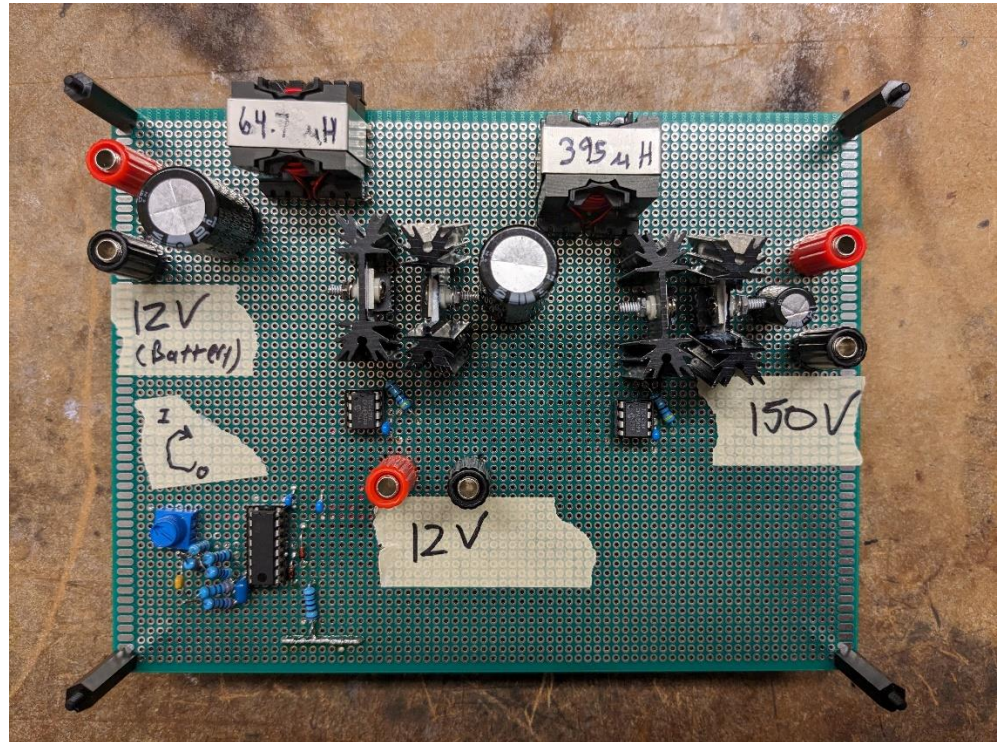


	Cursor 1	Cursor 2	Delta C2-C1
Frequency	100 kHz	10 kHz	-90 kHz
Trace 1	L_s	L_s	L_s
Measurement	64.742 μH	65.492 μH	750.332 nH
Trace 2	Phase ($^\circ$)	Phase ($^\circ$)	Phase ($^\circ$)
Measurement	89.358 $^\circ$	87.904 $^\circ$	-1.454 $^\circ$



	Cursor 1	Cursor 2	Delta C2-C1
Frequency	100 kHz	10 kHz	-90 kHz
Trace 1	L_s	L_s	L_s
Measurement	401.517 μH	395.723 μH	-5.794 μH
Trace 2	Phase ($^\circ$)	Phase ($^\circ$)	Phase ($^\circ$)
Measurement	89.789 $^\circ$	89.505 $^\circ$	-284.608 m $^\circ$

d. Cascaded Boost Power Stage Construction
i. Top and Bottom Side of Converter



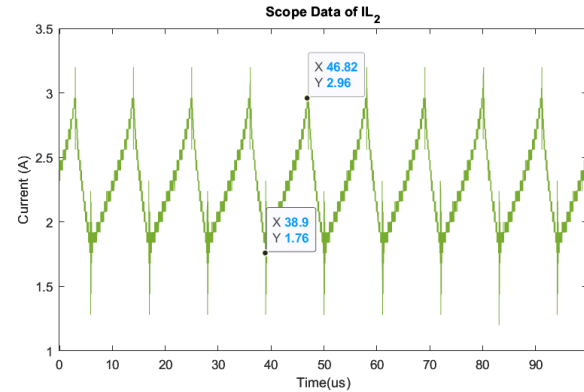
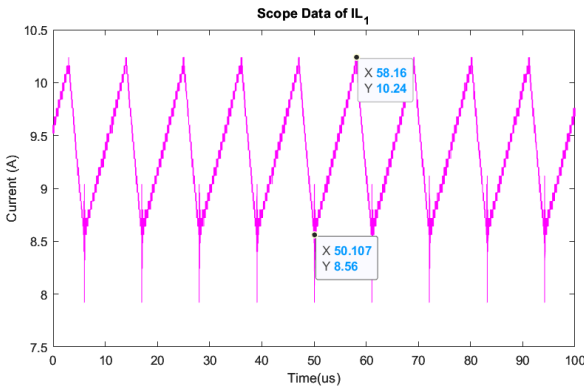
e. Cascaded Boost Converter Testing

i. Input and Output DC Voltages and Currents, Output Power, Efficiency

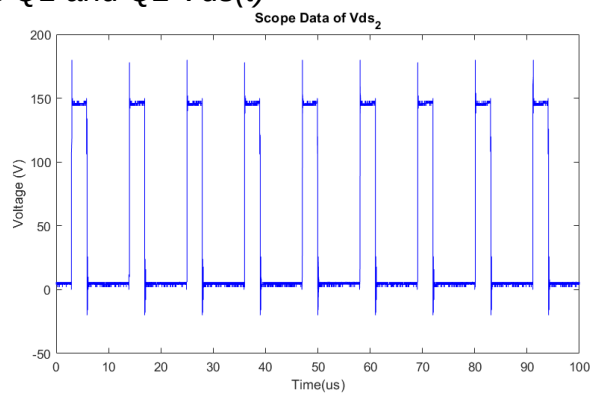
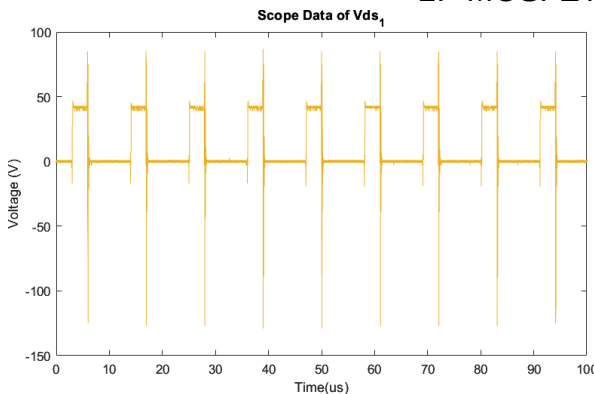
	Input Voltage	Input Current	Input Power	Output Voltage	Output Current	Output Power	Efficiency	Duty Cycle
Rload = 500Ω	11.99	4.15	49.7585	148.5	0.276	40.986	0.824	0.73
Rload = 262Ω	11.99	9.24	110.7876	149.8	0.578	86.5844	0.782	0.734

ii. Waveforms for Half-Load (85W) Power Out

1. Inductor Currents



2. MOSFETs Q1 and Q2 Vds(t)



iii. Measured Inductor Current Ripple

From the figures in II.e.ii.1, the following are calculated:

IL_1 : Min = 8.56A, Max = 10.24A, avg = 9.36A, **ripple = 0.84A**

This ripple is larger than the 0.7A we designed for but since this is a higher input power (110W) than was designed for (85W), it actually results in an overall smaller percentage ripple of only 9% compared to the design spec of 10%.

IL_2 : Min: 1.76A, max = 2.96A, avg = 2.29A, **ripple = 0.6A**

This ripple is considerably larger than the 0.3A, 15% ripple designed when 85W output power is experienced. This ripple is 26.2%. This may have been caused by the output voltage of the first converter not being at the expected 42.43V expected when calculating desired inductance for 15% ripple. It is also worth noting that the actual switching frequency was not 100 kHz, but rather 90 kHz. This, too, contributes to an inductor value that would've needed to be larger for the same desired 15%.

f. Loss Budget

i. Loss Budget Calculations and Documentation

There are 3 conduction losses to take into account from the small-signal-approximation waveforms: series resistance of inductor, R_{ds} of the transistor, and forward voltage of the diode. There are also 2 switching losses to account for: MOSFET switching loss and inductor core/proximity loss.

1. Waveform Conduction Losses

a. Series Resistance of Inductor

The average current through the first inductor was measured to be 9.36 A. Its DC series resistance was measured at 10.4 m Ω . Since $P = I_{rms}^2 R$, the copper conductor loss in this inductor is **0.911 W**. Similarly, with a resistance of 26.5 m Ω and an average current of 2.29 A, the second inductor's conductor loss is **0.139 W**. These DC losses will be used to calculate total copper loss after incorporating the skin effect in 2.b. below.

b. R_{ds} of Transistor

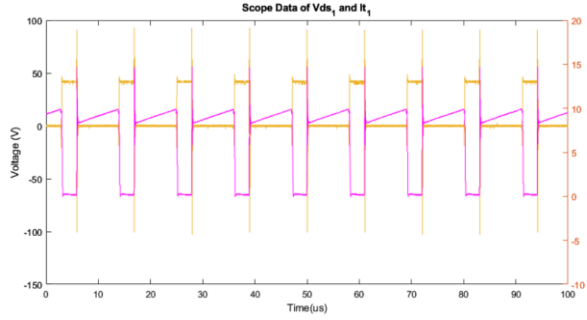
This calculation will assume inductor transistor current with ripple and use the RMS of the PWM signal of inductor current as shown that has amplitude average of 9.36A, ripple of 0.84A, and a duty cycle of $D = 0.734$. Thus, the RMS current through the first transistor is 8.03A. This corresponds to a conduction loss of $P = I_{rms}^2 R$ with $R_{ds(on)} = 11\text{m}\Omega$. Thus, 0.7093W is lost through conduction in the first MOSFET not including the spikes. The second MOSFET has an RMS current of 1.984A. This corresponds to a conduction loss of 2.087W when using an $R_{ds(on)}$ of 0.53 Ω . This data becomes irrelevant, however, by simply performing the average power calculation in 2.a. below and is not going to be included in the final power loss budget as a result.

c. Forward Voltage of Diode

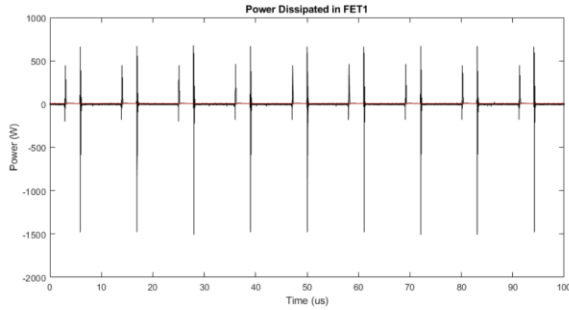
This power will be calculated using an RMS current through a diode that is of the form of a PWM with linear ripple. The RMS voltage will just be equivalent to the DC forward voltage of these diodes. For the first diode then, the RMS current is actually using the D' , or $D = 0.266$. Thus, I_{rms} of the first diode is 4.834A. It has a forward voltage of 0.95V, so the total power consumed would be **4.6W**. 1.19A is the I_{rms} in the second diode, resulting in a power consumption of **1.135W**.

2. Switching Losses

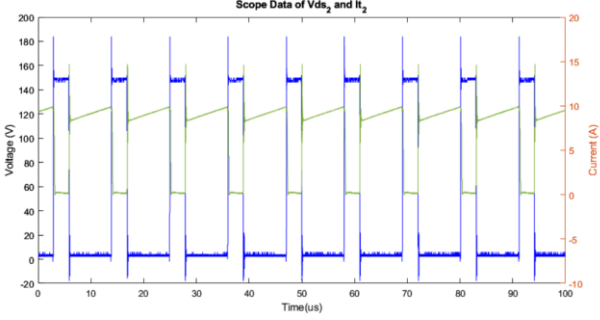
a. MOSFET Switching Loss



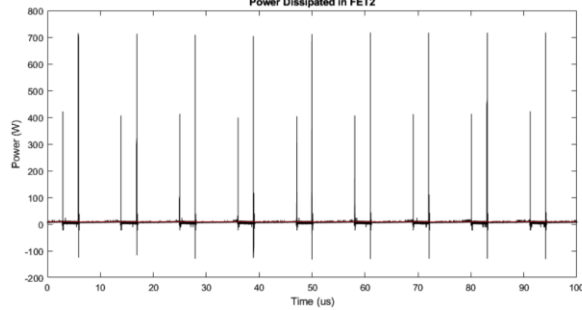
Yellow: Vds1, Magenta: IT1



Black = Power, Red = Averaged Power
FET 1 Averaged Power: **5.966W**



Blue: Vds2, Magenta: IT2



Black = Power, Red = Averaged Power
FET 2 Averaged Power: **9.389W**

b. Inductor Proximity/Core Loss

Skin Effect and Proximity Loss:

To calculate the proximity loss in the first inductor, we approximate the single layer of windings as a foil conductor of thickness h . Converting from the diameter of a 16 AWG

wire, we find a conductor thickness of $h = \sqrt{\frac{\pi}{4}} * 0.1291 = 0.1144$ cm. Next, we calculate

porosity from number of turns. The porosity is $\eta = \sqrt{\frac{\pi}{4}} * 0.1291 (\frac{13}{5.55}) = 0.268$ where 5.55cm

is the magnetic path length for the PQ26/25 core. From this approximation, we can use the following equations to determine the skin depth, assuming ambient temperature and a 90 kHz switching frequency.

$$\delta = \frac{\sqrt{\rho}}{\sqrt{\pi \mu_0 f}} = 0.025 \text{ cm}$$

$$\delta' = \frac{\delta}{\sqrt{\eta}} = 0.0483 \text{ cm}$$

Using the skin depth and the measured DC resistance for each of our respective inductors, we can calculate an AC resistance for each to use for the proximity effect.

$$R_{ac} = \frac{h}{\delta'} R_{dc}$$

This gives an AC resistance of 0.02463 Ω for the first inductor and 0.0399 Ω for the second which used 18 AWG.

Since the first inductor is just approximated using one layer, there is no need to overlay and observe the generation of eddy currents due to the proximity effect, and instead this loss is more of a high frequency skin effect loss alone. The Irms of just the ripple component of the inductor current that goes towards the edges is 0.485A. Thus, the skin effect loss in the first inductor is **5.8 mW**.

In the second inductor, the same process can be repeated to find the copper loss in the first layer, giving $P_1 = \left(\frac{I_{ripple}}{\sqrt{3}}\right)^2 R_{ac} = 4.8 \text{ mW}$. In the second layer, the proximity effect will induce eddy currents so that a current equal to the current through the inductor flows the opposite direction on one edge of the second layer. This means a current of twice this magnitude and opposite direction must flow on the other side of the layer to preserve the same net current through both layers. This results in a power of 5 times the power in the first layer being dissipated in the second layer, for a total A dissipation in the second inductor of **28.7 mW**.

Core Loss:

The $\Delta B = \mu n \Delta I$ for inductor 1 is $1980 \times 13 \times 0.84A = 0.027T$.

The $\Delta B = \mu n \Delta I$ for inductor 2 is $1980 \times 24 \times 0.60A = 0.036T$.

Core loss is expressed as $P_{fe} = K_{fe} \Delta B^\beta A_c l_m$, with a β of 2.6-2.8. Kfe is not a given value, so we will just use the N95 published data plots to approximate the volumetric loss given frequency and ΔB .

For inductor 1, the volumetric power density is 4.6 kW/m^2

For inductor 2, the volumetric power density is 8 kW/m^2

The $A_c l_m$ of this PQ26/25 core is $1.18 \times 5.55 = 6.549 \text{ cm}^2$

This results in core power loss in the first inductor of **3.013W**

In the second inductor, the power loss is **5.24W**

ii. Comparison to measured efficiency

Loss Source	Loss Amount (W)
Inductor DC Copper Loss	1.05
Diode Conduction Loss	5.735
Transistor Losses	15.355
Inductor AC Copper Loss	0.0345
Core Loss	8.253
Total:	30.4275

This is a much higher loss than we actually measured in our experiment, which was only 24.2 W of power lost. Possible sources of error include discrepancies in the reported datasheet loss elements from the realistic case (diode voltage drop, core loss),

as well as error in measurement tools. Measurement error seems especially likely due to our exorbitantly high switching losses.

iii. Efficiency Improvement Comments

Our efficiency could be improved by decreasing the switching frequency. Even with the measured switching loss being suspiciously high, it is still easily our largest source of inefficiency and the easiest to address.